Getting Started

From an empty workspace to a sliced, downloadable G-code in fifteen minutes. Then a map of where to go deeper on each subsystem.

ForgeSlicer is a **browser-based CAD + slicer combo**. You design with primitives + CSG, then slice with either the built-in JS engine or full server-side OrcaSlicer — no software install needed.

1. Workspace tour

Area	What's there	Shortcut
Left Panel	Primitives (Cube, Cylinder, Sphere, Bolt, Nut, Cone, Star, Sweep…), Combo (Slot, Fastener Pair, Countersink, Hex Pocket, Gusset), Library (Textures, Hardware, Components), Outliner.	—
Top Toolbar	Project menu, Slicer popover, Transform popovers (Move/Rotate/Scale/Mirror), Sketch toggle, Voice button, Help, Theme switcher.	—
3D Viewport	The build-plate scene. Orbit with right-mouse, pan with middle-mouse, zoom with wheel. Click any primitive to select.	—
Right Panel (Inspector)	Per-object properties: position, rotation, scale, dims, modifier, color, kind-specific tweakable fields.	—

2. Your first part — a name tag

End-to-end recipe to build, save, and download a personalised name tag in five minutes.

1. Sign in (if not already). Click **Workspace**. Empty build-plate appears.
2. From the **Primitives** tab, drop a **Cube**. In the Inspector set dims to **40 × 4 × 60 mm**. This is the tag body.
3. Drop another Cube. Set dims to **2 × 8 × 50 mm**. Position it at the top edge to act as a lanyard slot.
4. Set the second cube's **modifier** to **Negative** in the Inspector. The CSG engine carves it out of the first cube.
5. Drop a **Text** primitive (Primitives → Text). Type your name. Set height = 1.5 mm, position it on top of the tag.
6. Hit **Ctrl+S** to save the project. Name it «Name Tag».
7. Open the **Slicer** popover (Top Toolbar). Pick a printer (Custom → MyKlipper 0.4 nozzle is a fine default). Hit **Slice & Export GCODE**. The G-code preview shows the sliced result; click **Download .gcode**.

3. CSG — the engine behind everything

Every primitive in your scene carries a **modifier** flag: **positive** (added) or **negative** (subtracted). When the slicer runs:

1. All positive primitives are **union'd** into one solid.
2. All negative primitives are then **subtracted** from that solid.
3. The result becomes your STL, your 3MF, your G-code.

This is the single most important concept in ForgeSlicer. Most beginners try to draw the «final shape». Stop. Draw the **biggest shape** first, then carve away the bits you don't want with negative primitives. CSG = subtractive thinking.

4. Where to go next

Once you've made one part end-to-end, branch out into the deep-dive tutorials:

When you want to...	Read this	Time
Add gripable surfaces, knurls, hex grids, engraved logos	**Texture Library Tutorial**	20 min
Drop real M3–M12 or 1/4-20 bolt+nut assemblies into the scene	**Hardware Library Tutorial**	15 min
Build curved geometry — helical springs, arched handles, custom hooks	**Sweep + Sketch Tutorial**	25 min
Compare your slicer against full OrcaSlicer	Slicer popover → Compare engines (in-app)	5 min
Save and share your designs publicly	Gallery → Share (in-app)	—
Voice-command everything	Top Toolbar → Voice button (Shift+M)	—

5. Saving, sharing, and remixing

ForgeSlicer keeps your designs in three places:

- **Local project file** — Ctrl+S saves a JSON-encoded project the browser remembers. Closes-and-reopens cleanly.
- **Personal components** — right-click any object → **Save to library**. Reusable across all your projects.
- **Public gallery** — File → Share. Your design goes to the community gallery with a thumbnail, dimensions, and a «Remix» button so others can build on it. Cards now show extents in mm and an amber **too big** warning if a design exceeds the viewer's printer bed.

Remix is your friend. Public designs offer a green *Remix* button (or an amber *Remix · fit bed* when the model exceeds your printer). Hitting it opens the design in your workspace as the starting point for your iteration.

6. Built-in vs OrcaSlicer — which to pick?

Trait	Built-in (JS)	OrcaSlicer
Where it runs	In your browser	On the ForgeSlicer server
Speed	<1 s for typical scenes	5–30 s typical
Quality	Single-perimeter, no supports	Multi-perimeter, supports, ironing, calibration
Materials	Generic	BBL / Custom / community presets
Best for	Quick iteration, design verification	Final print-ready G-code

You don't have to choose blindly. The Slicer popover has a **Compare engines** button that slices the same scene through both, then shows a side-by-side metrics table with trophy icons on the winning side per row.

7. Keyboard cheatsheet

Action	Shortcut
Save project	Ctrl+S
Undo / Redo	Ctrl+Z / Ctrl+Y
Duplicate selection	Ctrl+D
Delete selection	Del / Backspace
Open Texture Library	Shift+T
Open Hardware Library	Shift+H
Enter / leave Sketch mode	K
Toggle Voice mode	Shift+M
Frame / fit-to-view	F
Toggle build-plate grid	G

8. Quick FAQ

Q. Does ForgeSlicer work offline?
A. The design tools and built-in slicer work offline once the page loads. OrcaSlicer slicing needs an internet connection (it runs server-side). Save-to-Gallery also needs network.

Q. Can I import STL/OBJ/3MF?
A. Yes — File → Import (or drag onto the viewport). The mesh becomes a regular scene object you can rotate, scale, and CSG-combine with primitives.

Q. How do I send a sliced print directly to my printer?
A. After slicing, the Slicer popover offers a **Send to printer** button. Currently supports Klipper / Moonraker on your LAN.

Q. The viewport stutters when I add lots of textures.
A. Texture objects are real geometry, not visual decals. Hit **tileSize × 2** in the Inspector — 4× fewer triangles for the same visual effect. The slicer still resolves the texture's relief height correctly.

Tutorial generated May 29, 2026 from ForgeSlicer source. Run the corresponding `scripts/build_*_tutorial.py` *to regenerate after future updates.*